TATA 1MG | HR OPERATIONS ARCHITECTURE SPECIFICATION

# Case Resolution & Domain Policy Roadmap

3-Step Guided Resolution Workspace & Inbound @1mg.com Identity Automation

*Document Version 2.4 | Engineering & People Operations Operational Blueprint*

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|---|---|---|---|
| **TARGET PLATFORM:** | HR Flow 2.0 (Case 360 Workspace) | **LIFECYCLE MODEL:** | 3-Step Consolidated Stepper |
| **GOVERNANCE:** | Tier-1 Zero-Trust Corporate PII Safeguards | **DOMAIN FILTER:** | @1mg.com Trust / External Auto-Hold |

## SECTION 1 Executive Overview & Architecture Objectives

This architectural blueprint addresses two mission-critical requirements for 1MG People Operations:

**1. Inbound Domain & Identity Policy Automation:**
Guarantees corporate emails (@1mg.com) auto-resolve while personal emails (e.g. Gmail) without Employee ID are held & clarified.

**2. 3-Step Consolidated Case Resolution Stepper:**
Replaces bloated multi-stage wizards with 3 intuitive phases: (1) Details & Dossier -> (2) Audit Section -> (3) Response & Closure.

## SECTION 2 Inbound Email Domain & Employee ID Policy Matrix

| SENDER DOMAIN | EMP ID IN BODY? | SYSTEM AUTOMATION ACTION | INITIAL STATUS | AUTO-EMAIL |
|---|---|---|---|---|
| @1mg.com (Corporate) | Optional / Any | **Auto-match directory email. Pre-fill EMP code,** department & role. Trust corporate identity. | READY | Ack (Standard) |
| External (@gmail.com...) | **YES (Extracted)** | **Capture EMP code. Match against employee DB** Stage for Specialist 360 Verification. | READY | Ack (Standard) |
| External (@gmail.com...) | **MISSING (Null)** | **Block processing! Auto-trigger Clarification Email** Require Employee ID reply to unlock case. | ON_HOLD | **AUTO-CLARIFY** |

**AUTOMATED CLARIFICATION DISPATCH TEMPLATE**

**Subject: Action Required: Please confirm your Employee ID - Case #{case_number}**

Dear {sender_name or 'Employee'},
We received your HR inquiry from a personal email address ({sender_email}). Under 1MG Data Security Policy, employee verification is required before case processing can begin.

**Please reply directly to this email with: (1) Official 1MG Employee ID (e.g. EMP1024), and (2) Full Name & Department.**

*Your case will unlock automatically upon reply. -- 1MG People Operations Desk*

**Key Principle: Zero-friction processing for verified @1mg.com corporate staff, combined with strict anti-spoofing** PII safeguards for external/personal email accounts.

SECTION 3

# Consolidated 3-Step Guided Resolution Lifecycle

Streamlining the Specialist Workbench from 4 fragmented tabs into 3 operational, milestone-driven phases:

**STEP 1: DETAILS & DOSSIER**
Inspect inbound mail & employee identity

→

**STEP 2: AUDIT SECTION**
Policy tasks, HRMS tools, ledger proof

→

**STEP 3: RESPONSE & CLOSURE**
Smart reply dispatch & case resolved

## STEP 1 Details & Dossier - Inbound Request & Identity Verification

**Core Specialist Capabilities:**

- Crisp Executive Letter Canvas: Formatted email body with quoted typography and TLS/SPF security telemetry.
- Employee 360 Identity Profile: Detects if sender is verified in DB or parsed from email fallback.
- Action Buttons: [Quick Add as Parsed] and [Review, Edit & Add to Database] to update SQLite master table.
- AI Entity Extraction: Automatic capture of EMP Code, dates, amounts, intent classification, and SLA deadline.
- Document Scanning: Verifies attached pay slips, medical certificates, or hospital prescriptions.
- Reassign Task Action: [Reassign Task] button in hero banner to handover case/task to specialized desk anytime.

*Outcome: Full case context established. Case is verified and ready for deep investigation in Step 2.*

## STEP 2 Audit Section - HR Investigation, Policy Checks & Forensic Ledger Proof

**Core Specialist Capabilities:**

- Policy Verification Checklist: Interactive tasks (Timesheet verification, Tax calculation audit, Leave quota check).
- Integrated HRMS Diagnostic Tools: Live checks (PAYROLL_TDS_CALC, LEAVE_QUOTA_AUDIT, PUNCH_SYNC).
- Manager Sign-Off Gate: Integrated approval workflow for high-risk payroll arrears or policy deviations.
- **Forensic Resolution Proof Entry (CRITICAL): Specialist enters tangible proof before closure:**
  - Outcome Type: SALARY_CREDITED, ARREARS_SCHEDULED, ATTENDANCE_REGULARIZED, etc.
  - Transaction Reference / Proof ID: Bank UTR Number (e.g. UTR-202610-849201), Batch ID, or Ledger ID.
  - Adjustment Amount: e.g. INR 14,500 (automatically linked into resolution records).
  - Audit Summary Notes: Detailed findings recorded permanently in the employee's HR profile.

*Outcome: Investigation completed with verifiable proof. Automatically prepares smart reply for Step 3.*

## STEP 3 Response & Resolution - Smart Employee Communication & Closure

**Core Specialist Capabilities:**

- Auto-Populated AI Resolution Composer: Pulls verified UTR proof, adjustment amount, and findings from Step 2.
- Template Presets: Quick-toggle between RESOLVE, CLARIFY / REQUEST INFO, STATUS UPDATE, or CUSTOM reply.
- Official Gmail SMTP Dispatch: Sends authenticated reply to employee from official PeopleDesk mailbox.
- Dual Resolution Actions: [Send Reply Only] (keeps case active) vs [Resolve Query & Dispatch] (closes case).
- Status Transition: Moves case to RESOLVED, stops SLA timers, and generates forensic audit log entry.
- Leave In Progress Integration: Specialist can pause and save draft at any time to resume later.

*Outcome: Employee receives formal resolution proof. Case closed cleanly with 100% forensic audit compliance.*

SECTION 4

# Implementation Phases & Technical Roadmap

## PHASE 1: BACKEND DOMAIN AUTOMATION ENGINE

**Component: backend/services/cases/case_engine.py & backend/services/gmail/gmail_fetcher.py**

- Step 1.1: Implement is_internal_company_domain(email) helper checking for @1mg.com domain suffix.
- Step 1.2: For @1mg.com senders: automatically look up SQLite employees table, auto-populate EMP code, and set READY.
- Step 1.3: For external senders WITHOUT Employee ID in body: set status=ON_HOLD, sub_status=AWAITING_EMP_ID.
- Step 1.4: Automatically dispatch SMTP clarification email requesting 1MG Employee ID and department.
- Step 1.5: Thread reply hook: when employee replies with their ID, automatically extract code, unlock case, and advance.

*Deliverables: Zero unverified cases processed; corporate staff auto-routed without manual data entry.*

## PHASE 2: FRONTEND 3-STEP STEPPER & AUDIT SECTION CONSOLIDATION

**Component: frontend/casedetail.html (UI & Interaction Controllers)**

- Step 2.1: Consolidate wizard-stepper-strip from 4 tabs to 3: (1) Details -> (2) Audit Section -> (3) Response.
- Step 2.2: Build unified Audit Section (renderStep2AuditHtml) containing Policy tasks, HRMS Tools, and Approval Gate.
- Step 2.3: Integrate Forensic Proof Input directly into Audit section: outcome dropdown, Bank UTR field, adjustment amount.
- Step 2.4: Build Step 3 Response View (renderStep3ResponseHtml) with auto-filled UTR findings and dual action buttons.
- Step 2.5: Retain Reassign Task & Leave In Progress headers for full operational flexibility.

*Deliverables: Ultra-fast 3-click case resolution workflow with forensic audit proof guaranteed.*

## PHASE 3: VERIFICATION, REGRESSION TESTS & DEPLOYMENT

**Component: backend/tests/ & End-to-End Test Suite**

- Step 3.1: Add unit tests verifying @1mg.com corporate sender auto-fill behavior.
- Step 3.2: Add unit tests verifying external domain (@gmail.com) without EMP ID triggers auto-reply and ON_HOLD.
- Step 3.3: Verify end-to-end case resolution on CASE-000021 through Steps 1 -> 2 -> 3 with UTR ledger proof.
- Step 3.4: Run full regression pytest suite ensuring 100% pass rate across existing 43 test cases.

*Deliverables: Zero regressions, fully automated audit trail, robust multi-domain security.*

## ARCHITECTURAL SPECIFICATION APPROVAL & SUMMARY

`Document ID: SPEC-HRFLOW-2026-V2.4`

Lead Architect: Google DeepMind Advanced Agentic Systems

Stakeholder Desk: Tata 1mg People Operations (Payroll, Leave & General HR)

Database Backend: SQLite (hrflow.db) | employees, cases, emails, audit_logs tables

**Resolution Standard: Bank UTR / HRMS Transaction Reference required for all payroll discrepancies.**

**Status: Approved for Implementation | Ready to proceed with Phase 1 Backend Automation.**